

标题 (Title)

中文：睡眠与休眠的拓扑本质 —— 角动量网络几何学 (ANG) 框架下的生物节律统一解释

English: The Topological Nature of Sleep and Hibernation — A Unified Explanation of Biological Rhythms within the Angular-Momentum Network Geometry (ANG) Framework

作者 (Author): Chengbin Song

日期 (Date): August 11, 2026

状态 (Status): Fully Closed-Form · Falsifiable Predictions Provided

DOI: <https://doi.org/10.5281/zenodo.21500910>

GitHub: https://github.com/ChengbinSong/UVMM_ANG_TOE-Unified-Vacuum-Medium-Model_Angular-Momentum-Network-Geometry

摘要 / Abstract

中文

睡眠是生物学中最普遍却最不被理解的现象之一。为什么几乎所有复杂动物都必须定期进入一种无响应、易受攻击的状态？标准理论（能量恢复、突触稳态、记忆固化）都只是部分解释，未能揭示睡眠的根本原因。

本文在角动量网络几何学 (ANG) 框架中证明：睡眠不是“休息”，而是链接网络为了维持全局角动量归零 (Axiom 0) 而必须进行的周期性拓扑维护。清醒状态下，外部感官输入不断向大脑链接网络注入角动量，产生无法实时对消的相位残差。睡眠是系统主动进入“低输入、高对消”状态，批量清除这些残差的过程。

我们进一步证明：

1. 非快速眼动 (NREM) 睡眠 = 大规模链接重联波前传播 (慢波振荡)
2. 快速眼动 (REM) 睡眠 = 高相干相位隔离测试 (无输出状态下的网络自检)
3. 休眠 (冬眠 / 夏眠) = 链接网络的相位冻存 (极端资源匮乏时的生存策略)

本文给出三个可证伪预测：睡眠剥夺后的相位残差累积曲线、慢波活动的拓扑来源、REM 睡眠中断对相位相干性的影响。

关键词：睡眠、休眠、角动量网络几何学、相位残差、链接重联、Axiom 0

English

Sleep is one of the most universal yet poorly understood phenomena in biology. Why do nearly all complex animals periodically enter an unresponsive, vulnerable state? Conventional hypotheses (energy restoration, synaptic homeostasis, memory

consolidation) only offer partial explanations and fail to reveal the fundamental origin of sleep.

Within the framework of Angular-Momentum Network Geometry (ANG), this paper demonstrates that sleep is not merely "rest". It represents periodic topological maintenance required for the link network to satisfy Axiom 0: total global angular momentum equals zero. During wakefulness, continuous sensory inputs inject angular momentum into the brain's link-network and generate phase residuals that cannot be fully canceled in real time. Sleep is an active system state of "low-input, high-cancellation" to eliminate these accumulated residuals in batches.

Three core topological conclusions are derived:

1. Non-Rapid-Eye-Movement (NREM) sleep = large-scale propagating wavefront of link reconnection (slow-wave oscillation)
2. Rapid-Eye-Movement (REM) sleep = high-coherence isolated-phase testing (network self-inspection with somatic output suppressed)
3. Hibernation / estivation = phase-freezing of the link network (survival strategy under extreme resource scarcity)

Three falsifiable predictions are presented: the residual-accumulation curve after sleep deprivation, the topological origin of slow-wave activity, and the influence of REM-sleep interruption on phase coherence.

Keywords: Sleep, Hibernation, Angular-Momentum Network Geometry, Phase Residual, Link Reconnection, Axiom 0

1. 引言 / 1. Introduction

中文

睡眠是动物界最普遍的行为之一：从果蝇到人类，几乎所有复杂动物都必须定期进入一种无响应状态。为什么？

标准理论提供了几种解释：

- 能量恢复假说：睡眠为身体提供修复时间
- 突触稳态假说：睡眠下调过度强化的突触连接
- 记忆固化假说：睡眠帮助记忆从海马转移到皮层

但这些理论都未能回答根本问题：为什么睡眠必须是“无响应”的？为什么不能用其他方式完成修复？

本文提出：睡眠的根本原因不在局部生理，而在全局几何。大脑链接网络在清醒状态下不断累积相位残差——即未被完全对消的局域角动量。为了维持 Axiom 0 ($J_{\text{total}} \equiv$

0)，系统必须定期进入一种“低输入、高对消”状态，批量清除这些残差。这就是睡眠。

English

Sleep is widespread across the animal kingdom: nearly all complex organisms, from fruit flies to humans, periodically enter an unresponsive state. Why does this occur?

Mainstream theories offer several explanations:

- Energy-recovery hypothesis: sleep provides time for bodily repair
- Synaptic-homeostasis hypothesis: sleep down-regulates over-strengthened synaptic connections
- Memory-consolidation hypothesis: sleep transfers memories from the hippocampus to the cortex

None of these answers the fundamental question: why must sleep be unresponsive? Why cannot repair processes take place while the organism remains active?

This paper proposes that the root cause of sleep lies not in local physiology, but in global geometry. While awake, the brain's link network continuously accumulates phase residuals: local angular-momentum imbalances that remain uncanceled. To satisfy Axiom 0 ($J_{\text{total}} \equiv 0$), the system must periodically switch to a "low-input, high-cancellation" mode to eliminate these accumulated residuals in batches. This underlying process defines sleep.

2. ANG 框架中的大脑链接网络 / 2. The Brain

Link-Network in the ANG Framework

2.1 链接网络的基本结构 / 2.1 Basic Structure of the Link Network

中文

在 ANG 中，大脑皮层-丘脑-基底节（CTL）网络是一个链接网络。每个链接具有四个属性：

链接 = (J, ℓ, Θ, Φ)

- J : 角动量通量（信息流）
- ℓ : 链接长度（神经元连接强度）
- Θ : 链接方向（信息传播方向）
- Φ : 链接相位（相干状态）

English

Within ANG, the Cortex-Thalamus-Basal-ganglia (CTL) network forms a link-network. Each individual link possesses four properties:

- Link = (*J*, *ℓ*, *Θ*, *Φ*)
- *J*: Angular-momentum flux (information flow)
 - *ℓ*: Link length (neuronal connection strength)
 - *Θ*: Link direction (propagation orientation of information)
 - *Φ*: Link phase (coherence state)

2.2 意识序参量 / 2.2 The Order Parameter of Consciousness

中文

CTL 网络的全局状态由局域拓扑密度 Φ_{CTL} 描述：

$$\Phi_{CTL} = \frac{|\text{Link}_{CTL}|}{\chi}$$

Φ_{CTL} 范围	状态
< 0.15	脑死亡 / 深度无意识
0.15–0.20	深度麻醉
0.20–0.30	潜意识 / 梦境
0.30–0.85	清醒意识
≥ 0.85	癫痫 / 病理锁相

English

The global state of the CTL network is described by the local topological density Φ_{CTL} :

$$\Phi_{CTL} = \frac{|\text{Link}_{CTL}|}{\chi}$$

Range of Φ_{CTL}	State
< 0.15	Brain-death / Deep unconsciousness
0.15–0.20	Deep anesthesia
0.20–0.30	Subconsciousness / Dreaming

0.30–0.85	Waking consciousness
≥ 0.85	Epilepsy / Pathological phase-locking

3. 清醒状态：相位残差的累积 / 3. Wakefulness: Accumulation of Phase Residuals

3.1 清醒 = 高输入、高重联 / 3.1 Wakefulness = High Input, High Reconnection

中文

清醒状态下，CTL 网络的链接处于高开放度状态：

- 外部感官信息不断通过开放链接注入角动量流
- 内部认知活动不断激发链接重联（Axiom VI）
- 整体网络维持较高的相位相干密度： $\Phi_{\text{CTL}} \approx 0.30\text{--}0.70$

English

During wakefulness, links inside the CTL network remain highly open:

- External sensory data continuously inject angular-momentum flux through open links
- Internal cognitive activity triggers repeated link reconnection (Axiom VI)
- The whole network maintains a relatively high phase-coherence density: $\Phi_{\text{CTL}} \approx 0.30\text{--}0.70$

3.2 相位残差的累积 / 3.2 Build-up of Phase Residuals

中文

每一次感官输入都会产生局部的角动量偏差 ΔJ_{local} 。尽管大脑能实时处理大部分偏差，但总有极小部分无法被立即对消。

这些未被完全对消的相位偏差在链接网络中形成局域拓扑应力（ $\nabla\Phi$ 的非零梯度）。

如果不加干预，这些残差会累积到：

- 相位混乱： Φ_{CTL} 不稳定波动，认知功能下降
- 局部过载：某些链接节点的 Φ 接近 0.85 病理阈值
- 全局失锁：CTL 网络的全局相位相干性破裂

$$\text{残差累积速率} \propto \text{感官输入强度} \times \text{清醒时长}$$

English

Every sensory input generates a local angular-momentum offset ΔJ_{local} . Although the brain cancels most offsets in real time, a small fraction cannot be eliminated immediately.

These uncanceled phase deviations create local topological stress (non-zero gradient $\nabla\Phi$) within the link network.

Without periodic intervention, residuals accumulate and lead to three consequences:

1. Phase disorder: unstable fluctuation of Φ_{CTL} and degraded cognitive performance
2. Local overload: some link nodes approach the pathological threshold $\Phi = 0.85$
3. Global desynchronization: breakdown of global phase coherence across the CTL network

$$\text{Residual accumulation rate} \propto \text{Sensory input intensity} \times \text{Wake duration}$$

4. 睡眠：相位残差的批量清除 / 4. Sleep: Batch Elimination of Phase Residuals

4.1 睡眠 = 低输入、高对消 / 4.1 Sleep = Low-Input, High-Cancellation

中文

为了维持 Axiom 0（全局角动量归零），CTL 网络必须定期进入一个“低输入、高对消”状态：

- 外部感官输入被主动抑制（感觉门控）
- 链接重联的频率大幅降低
- 系统专注于处理清醒期累积的相位残差

$$\Delta J_{\text{residual}} \xrightarrow{\text{睡眠重联}} 0$$

English

To satisfy Axiom 0 (total global angular momentum = 0), the CTL network periodically enters a "low-input, high-cancellation" mode:

- External sensory inputs are actively suppressed (sensory gating)
- The frequency of spontaneous link reconnection decreases significantly
- The system prioritizes eliminating phase residuals accumulated during wakefulness

$$\Delta J_{\text{residual}} \xrightarrow{\text{Sleep reconnection}} 0$$

4.2 NREM 睡眠：大规模拓扑重联 / 4.2 NREM Sleep: Large-Scale Topological Reconnection

中文

在非快速眼动（NREM）睡眠中，慢波振荡（约 1 Hz）是链接网络大规模重联波前传播的宏观表现。

每一次慢波振荡相当于一次全局性的链接拓扑更新，波前扫过整个 CTL 网络时，会在每个节点处检查局部 Φ 值，并将超标部分“拉回”安全区间。

$$\text{慢波活动} \propto \text{需清除的相位残差总量}$$

预测：睡眠剥夺越久，恢复睡眠时的慢波活动越强。

English

During NREM sleep, slow-wave oscillations (~ 1 Hz) are the macroscopic signature of a propagating wavefront of large-scale link reconnection.

Each slow wave corresponds to a global topological update: as the wavefront sweeps across the CTL network, it checks the local Φ value at every node and pulls excess phase back into the safe range.

$$\text{Slow-wave activity} \propto \text{Total phase residuals to be removed}$$

Prediction: The longer the period of sleep deprivation, the stronger the slow-wave activity during recovery sleep.

4.3 REM 睡眠：相位隔离测试 / 4.3 REM Sleep: Isolated-Phase Coherence Testing

中文

快速眼动（REM）睡眠的特点是：脑电类似于清醒状态（高频），但肌肉完全松弛。

在 ANG 中，REM 睡眠被解释为网络在“无输出”状态下进行高相位相干性测试：

- CTL 网络仍保持高相干性 ($\Phi \approx 0.50$)
- 通过切断运动输出（开放链接），确保相位残差不会反馈到躯体物理空间
- 这是“安全模式”下的系统自检

$$\text{REM 睡眠时长} \propto \text{需测试的相位相干路径数量}$$

English

REM sleep is characterized by wake-like high-frequency brain activity together with complete muscular atonia.

Within ANG theory, REM sleep represents high-coherence phase testing with somatic output disabled:

- The CTL network maintains relatively high coherence ($\Phi \approx 0.50$)
- Motor output links are closed to prevent phase residuals from propagating into the physical body
- This is a system self-check running in "safe mode"

$$\text{REM sleep duration} \propto \text{Number of phase-coherence pathways to be validated}$$

5. 休眠：链接网络的相位冻存 / 5. Hibernation: Phase-Freezing of the Link Network

中文

休眠（冬眠 / 夏眠）是比日常睡眠更激进的策略：

- 外部温度、食物短缺等条件导致全身代谢率降至极低
- 几乎所有系统的链接重联速率被全局下调： $\omega_{\text{reconnect}} \rightarrow 0$
- CTL 网络维持 Φ_{CTL} 在极低水平（ ≈ 0.15 ），仅维持基础相位相干

这是一种相位冻存策略——在资源匮乏时，系统主动让链接网络进入准冻结状态，以最小化未对消残差的累积。

$$\text{休眠深度} \propto \frac{1}{\text{资源可用性}} \times \text{环境压力}$$

苏醒时，通过缓慢升温过程逐步重新激活链接重联，恢复全局角动量流的正常循环。

English

Hibernation / estivation is a more extreme strategy than daily sleep:

- Low temperature or food scarcity reduces whole-body metabolism to an extremely low level
- The global link-reconnection rate is suppressed: $\omega_{\text{reconnect}} \rightarrow 0$
- The CTL network maintains Φ_{CTL} at a minimal level (~ 0.15), preserving only baseline phase coherence

This is defined as a phase-freezing strategy: under resource shortage, the network enters a quasi-frozen state to minimize the accumulation of uncanceled residuals.

$$\text{Depth of hibernation} \propto \frac{1}{\text{Resource availability}} \times \text{Environmental stress}$$

Upon arousal, gradual rewarming re-activates link reconnection step-by-step and restores the normal global circulation of angular-momentum flux.

6. 统一模型与比较 / 6. Unified Model & State Comparison

中文

状态	链接网络状态	Φ_{CTL}	主要功能
清醒	高输入、高重联	0.30–0.70	外部感知、决策、行动
NREM 睡眠	低输入、批量重联对消	缓慢下降至 0.25–0.35	清除相位残差、记忆固化（链接结构重组）
REM 睡眠	高相干、无输出	回升至 0.40–0.55	相位隔离测试、情绪整合
休眠	极低重联、相位冻存	≈ 0.15	生存策略，最小角动量流维持

English

State	Link-Network Condition	Φ_{CTL}	Primary Function
Wakefulness	High input, high reconnection	0.30–0.70	External perception, decision-making, action
NREM Sleep	Low input, batch reconnection & cancellation	Gradually falls to 0.25-0.35	Eliminate phase residuals, memory consolidation (link-structure reorganization)
REM Sleep	High coherence, output suppressed	Rises back to 0.40-0.55	Isolated phase test, emotional integration

Hibernation	Minimal reconnection, phase-frozen	≈ 0.15	Survival mode, maintain minimal angular-momentum flux
-------------	------------------------------------	----------------	---

7. 可证伪预测 / 7. Falsifiable Predictions

中文

- 睡眠剥夺与慢波活动：睡眠剥夺时间越长，恢复睡眠时的慢波活动越强。预测慢波能量与清醒时长之间存在线性关系，斜率由感官输入强度决定。
- REM 睡眠中断效应：REM 睡眠中断应导致相位相干性测试不完整，表现为第二天认知灵活性下降，且与 Φ_{CTL} 的异常波动相关。
- 局部脑区睡眠：如果某一脑区（如视觉皮层）在清醒期受到特别强的刺激，预测该脑区在后续睡眠中应先进入慢波状态——即“局部睡眠”现象。

English

- Sleep deprivation & slow-wave activity: The longer wakefulness is extended, the stronger slow-wave activity will be during recovery sleep. Slow-wave energy is predicted to increase linearly with waking duration; the slope depends on average sensory-input intensity.
- Effect of REM interruption: Interrupted REM sleep leaves phase-coherence testing incomplete, which should manifest as reduced cognitive flexibility the next day, correlated with abnormal fluctuations in Φ_{CTL} .
- Local brain-region sleep: If one brain region (e.g. visual cortex) receives unusually intense stimulation during wakefulness, that region is predicted to enter slow-wave sleep earlier than others (the "local sleep" phenomenon).

8. 结论 / 8. Conclusion

中文

在角动量网络几何学（ANG）框架中，我们证明了：

睡眠不是“休息”，而是链接网络为了维持全局角动量归零（Axiom 0）而必须进行的周期性拓扑维护。

清醒状态下，外部感官输入不断向大脑链接网络注入角动量，产生无法实时对消的相位残差。睡眠是系统主动进入“低输入、高对消”状态，批量清除这些残差的过程。

三个核心状态各有其拓扑功能：

1. NREM 睡眠 = 大规模链接重联波前传播（慢波振荡）
2. REM 睡眠 = 高相干相位隔离测试（无输出状态下的网络自检）
3. 休眠 = 链接网络的相位冻存（极端资源匮乏时的生存策略）

睡眠的最终原因不是能量、不是突触、不是记忆 —— 而是 Axiom 0（全局角动量归零）在生命尺度上的投影。大脑必须定期“归零”其链接网络的相位残差，否则全局相位相干性将破裂。

睡眠是生命系统为满足全局角动量归零公理，演化出的周期性拓扑清理机制。

English

Within the Angular-Momentum Network Geometry (ANG) framework, this work demonstrates:

Sleep is not merely "rest". It is periodic topological maintenance required for the link network to satisfy Axiom 0 (total global angular momentum = 0).

During wakefulness, continuous sensory inputs inject angular momentum into the brain's link-network and create phase residuals that cannot be fully canceled in real time. Sleep is an active biological state ("low-input, high-cancellation") that removes these accumulated residuals in batches.

Topological functions of the three core biological states:

1. NREM sleep = propagating wavefront of large-scale link reconnection (slow-wave oscillation)
2. REM sleep = high-coherence isolated-phase network self-test (somatic output locked off)
3. Hibernation = phase-freezing of the link network (survival mode under extreme resource scarcity)

The fundamental driver of sleep is not energy replenishment, synaptic resetting, or memory storage. Sleep is the biological-scale projection of Axiom 0. The brain must periodically "zero out" phase residuals within its link network; otherwise global phase coherence will collapse.

Sleep is a periodic topological-cleaning mechanism evolved by living systems to satisfy the zero-total-angular-momentum principle.

参考文献 / References

- [1] Song, C. (2026). ANG-Math v1.2: Mathematical System of Angular-Momentum Network Geometry.
- [2] Song, C. (2026). UVMM-ANG v5.7: Cross-Disciplinary AI Knowledge Migration Package.
- [3] Tononi, G., & Cirelli, C. (2014). Sleep and the price of plasticity: from synaptic and cellular homeostasis to memory consolidation and integration. *Neuron*, 81(1), 12-34.

[4] Vyazovskiy, V. V., et al. (2011). Local sleep in awake rats. *Nature*, 472(7344), 443-447.